

Submission Cover Sheet

Course Details

Title: Applied Data Science

Code: DAT5003 S2_24

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Assessment: **WRIT1, Task3 – Analytical Summary Report**

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Analytical Summary Report

The Data

The dataset provides a year of average hourly readings from a metal oxide semiconductor pollution sensor in an Italian city, recorded in 2004 to 2005. There are readings from several sensors, each tuned to a particular pollutant. Readings were also taken for temperature and humidity.

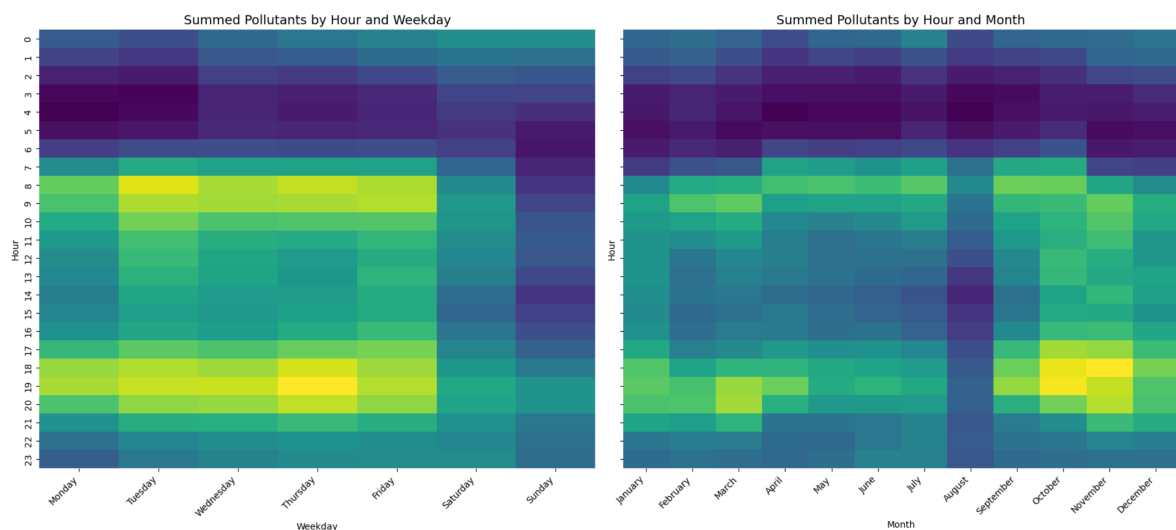
Analytical Questions

This report aims to answer the following questions:

- Is it possible to predict future pollution levels based on historical data?
- Can this data be used to inform residents of the city about potentially risky times, and when to take precautions or to avoid going outside without a mask?

Summary of Findings

These heatmaps show that pollution levels are at their highest from 0730 to 1000, and from 1630-2030 on weekdays

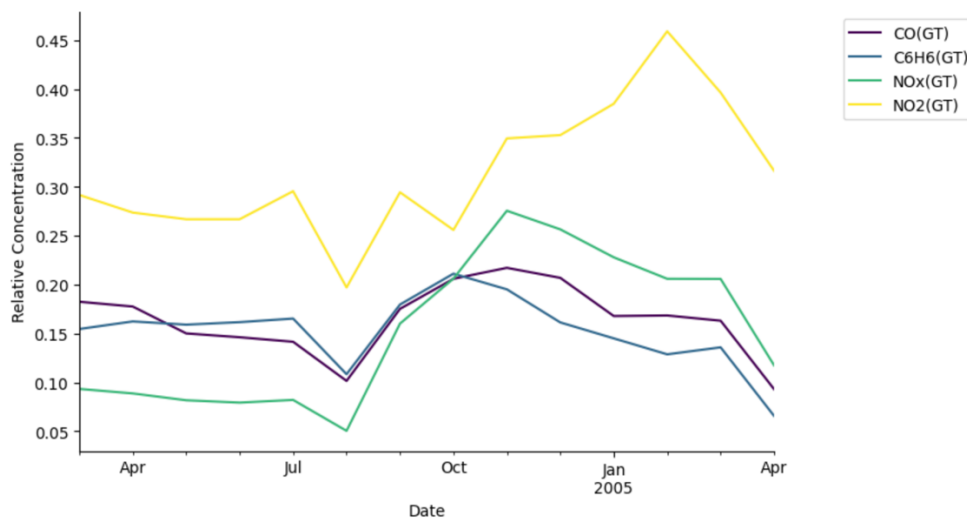


The pollution is present throughout daytime hours. Thursday evenings are the worst time to be outside. The air is less polluted at weekends, especially on Sunday.

This effect is greater winter months, when more people drive to work due to the colder, wetter weather. August is comparatively pollution free when Italians tend to take their summer holidays.

These charts are likely to be a good indicator of future pollution levels.

This chart shows the relative concentration of pollutants and identifies periods of high pollution over time, to allow focussed public health interventions:



Unfortunately, the pollutant levels frequently breached World Health Organisation safety limits:

Pollutant	24 Hour Average	1 Hour Average
CO (Carbon Monoxide)	8 times (2%)	0 times
C6H6 (Benzene)	381 times (98%)	94 times (1%)
NO2 (Nitrogen Dioxide)	391 times (100%)	400 times (4%)

Positive Actions

To reduce pollution levels, the following actions can be taken by city planners:

- Encourage home working
- Incentivise using public transport
- Incentivise using electric vehicles and bicycles
- Define 'low pollution zones' like London ULEZ

Conclusion

It is possible to predict future pollution levels from past data, particularly with reference to time of day and month of the year.

(290 words)